



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

BACHELOR OF COMPUTER SCIENCE (DATA ENGINEERING) WITH HONS.

FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

SECP3843 - SPECIAL TOPIC IN DATA ENGINEERING (WBL)

TUTORIAL ARTICLE

GROUP: 8

PREPARED BY :

NAME	MATRIC NO
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
NURUL ADRIANA BINTI KAMAL JEFRI	A23CS0258

PREPARED FOR:

DR. ARYATI BINTI BAKRI

TABLE OF CONTENTS

1.0 Main problem	3
2.0 Main solution	3
3.0 User story	4
3.1 Top Management	4
3.2 Operations / Production Manager	4
3.3 Sales Team	4
3.4 Data Analyst	5
3.5 Finance Department	5
4.0 ERD based on above case study	5
5.0 Use case diagram	8
6.0 Design Data/system architecture	9
6.1 Data Flow Summary	9
7.0 Multidimensional model	11
8.0 Conclusion	12

1.0 Main problem

The core business processes of PT. MPM did not have a proper system to measure and monitor the performance of the core business processes, more specifically;

- There was a lack of a fit-for-purpose application or system to effectively measure and monitor performance of operations, causing top management to make ineffective decisions.
- Data in various departments (Sales, Manufacturing, Inventory, etc.) were in disorganized, piecemeal forms without a single view.
- Without visibility of KPIs, the business had difficulties assessing actuals vs. forecasts, sales and production, and made themselves less competitive compared to the competitors.
- Performance was not measured in real-time or centrally or in an accessible manner (e.g., OEE, revenue, production capacity).

2.0 Main solution

Create a Business Intelligence (BI) Dashboard following the Waterfall approach and Google Data Studio that includes two dashboards;

Table 2.0.1 Proposed dashboards and its purpose

Dashboard	Purpose
Sales Performance Dashboard	Enable monitoring of revenue, forecasting comparisons, product categories and customer performance.
Production Performance Dashboard	Monitor overall equipment effectiveness (OEE), machine performance and planned vs actual production.

3.0 User story

The following user stories are derived from the system requirements and stakeholder needs identified in the research:

3.1 Top Management

1. As top management, I want to view the overall sales performance on a single dashboard, so that I can assess whether revenue and sales KPIs are being met without requesting manual reports.
2. As top management, I want to compare forecast versus actual sales and revenue by period, so that I can identify deviations and make timely strategic adjustments.
3. As top management, I want to monitor OEE (Overall Equipment Effectiveness) across all production lines, so that I can evaluate production efficiency and take corrective action.
4. As top management, I want to access the dashboard from any device with an internet connection, so that I can stay informed even when not on-site.

3.2 Operations / Production Manager

1. As a production manager, I want to drill down into individual machine performance metrics (Availability, Performance, Quality), so that I can pinpoint underperforming machines and allocate maintenance resources effectively.
2. As a production manager, I want to see a bar chart comparing planned vs. actual production output over time, so that I can identify production bottlenecks and address scheduling issues.
3. As a production manager, I want to filter production data by product division and machine, so that I can conduct focused performance reviews per department.

3.3 Sales Team

1. As a sales team member, I want to view sales performance broken down by product category and customer ID, so that I can identify high-performing and underperforming product lines.
2. As a sales team member, I want to see quarterly performance trends via a line chart, so that I can track seasonality patterns and adjust sales strategies accordingly.

3.4 Data Analyst

1. As a data analyst, I want to upload and update the data warehouse in Google Sheets, so that the dashboards always reflect the most current operational data.
2. As a data analyst, I want to perform ETL (Extract, Transform, Load) operations on raw report data, so that clean and structured data is consistently available for visualization.

3.5 Finance Department

1. As a finance team member, I want to view revenue data segmented by product and period, so that I can support financial forecasting and budget planning processes.

4.0 ERD based on above case study

Based on the article case study Development of Performance Monitoring Dashboard for Product Packaging Manufacturer by Jayadinata et al. (2024), the ERD is derived from the DFD level 0 of the paper. In the DFD level 0, there are six entities that interact with the ERP process, which leads to a self-service business intelligence system that delivers a dashboard to top management. As shown in Figure 4.0 is the ERD based on the case study.

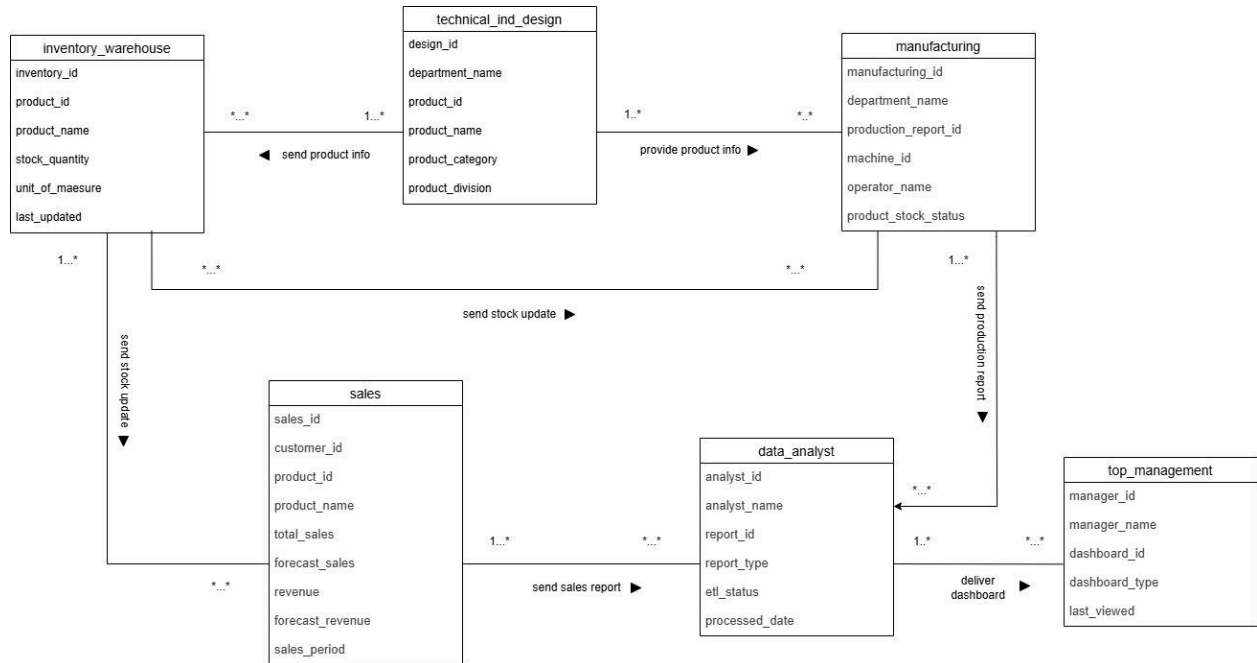


Figure 4.0 ERD based on the case study

Based on the derived ERD, below is entity list and its description workflow:

- **technical_ind_design**

Sends product info to the inventory_warehouse and provides product info to the manufacturing department.

- **manufacturing**

Received product specs from the design team. It sends raw production reports downstream to the data_analyst to monitor throughput and status, and exchanges or receives stock status updates directly with the warehouse.

- **inventory_warehouse**

Uses product details provided by the design team to initialize inventory records. Constantly broadcasts real-time stock balances (send stock update) to both the manufacturing floor and the sales team to prevent overselling or overproduction.

- **sales**

Captures transactional records, commercial metrics, and predictive forecasting and compiles commercial outcomes and transmits them to the data_analyst.

- **data_analyst**

Act as a central hub for downstream reporting. It received data from two different functional areas which are sales and manufacturing.

- **top_management**

After data_analyst delivers a finished BI dashboard (deliver dashboard) to management for tracking high-level KPIs

5.0 Use Case Diagram

Based on the article case study by Jayadinata et al. (2024), the use case diagram is drawn and named as the performance monitoring dashboard system. It is designed to integrate data from multiple departments within the organization to provide top management with real-time insight into operational performance. The use case diagram in Figure 5.1 shows the interaction between these actors and the system.

1. **Actors:** Technical & Ind. Design, Inventory & warehouse, Manufacturing, Sales/Logistic, Data Analyst, Top Management
2. **Use case diagram description:**
 - Actors interact with system processes that represent their responsibilities.
 - Each process corresponds to a use case to show what functionality the system provides
 - Data analyst interact with ETL process to prepare clean data warehouse
 - Top management interact with dashboard to monitor KPIs and make decisions
 - Sales, manufacturing, inventory, and design departments provide operational data inputs which flow into the BI system.

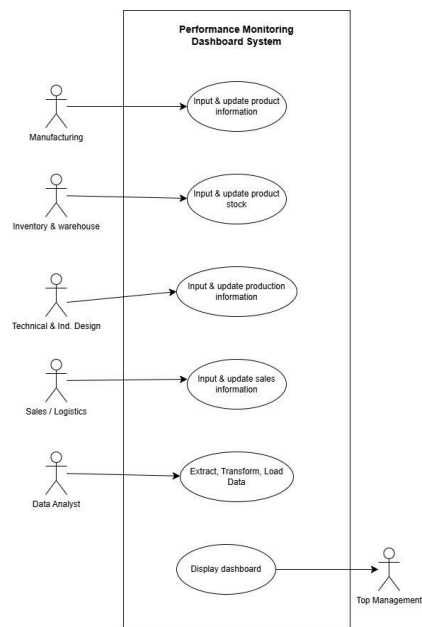


Figure 5.1 Use case diagram

6.0 Design Data/System Architecture

The data architecture of the PT. MPM Business Intelligence (BI) dashboard system follows a structured data warehouse model consisting of four main components: data sources, ETL process, data warehouse with data marts, and visualization layer as shown in Figure 6.0.

The first component is the data sources layer, which consists of five operational systems that generate raw data within the organization. These include the ERP system, Sales department, Manufacturing department, Inventory & Warehouse, and Technical & Industrial Design. Each of these sources contributes transactional and operational data that forms the basis of the BI system.

The raw data from all five sources is then channeled into the ETL (Extract, Transform, Load) process, which is carried out using Google Sheets. During this stage, data is extracted from the source systems, cleaned to remove redundant or irrelevant records, transformed into a consistent and structured format, and loaded into the central data warehouse.

The data warehouse serves as the central repository for all processed and structured data. From the data warehouse, data is further separated into two dedicated data marts, the Sales Data Mart, which stores sales-related dimensions such as revenue, forecast, product category, and customer information, and the Production Data Mart, which stores production-related dimensions including OEE (Overall Equipment Effectiveness), machine performance, availability, and quality metrics.

Finally, the visualization layer is powered by Google Data Studio, which connects directly to the data marts to render two interactive dashboards: the Sales Performance Dashboard and the Production Performance Dashboard. These dashboards provide top management with real-time insights and key performance indicators (KPIs) to support data-driven decision-making and business strategy development.

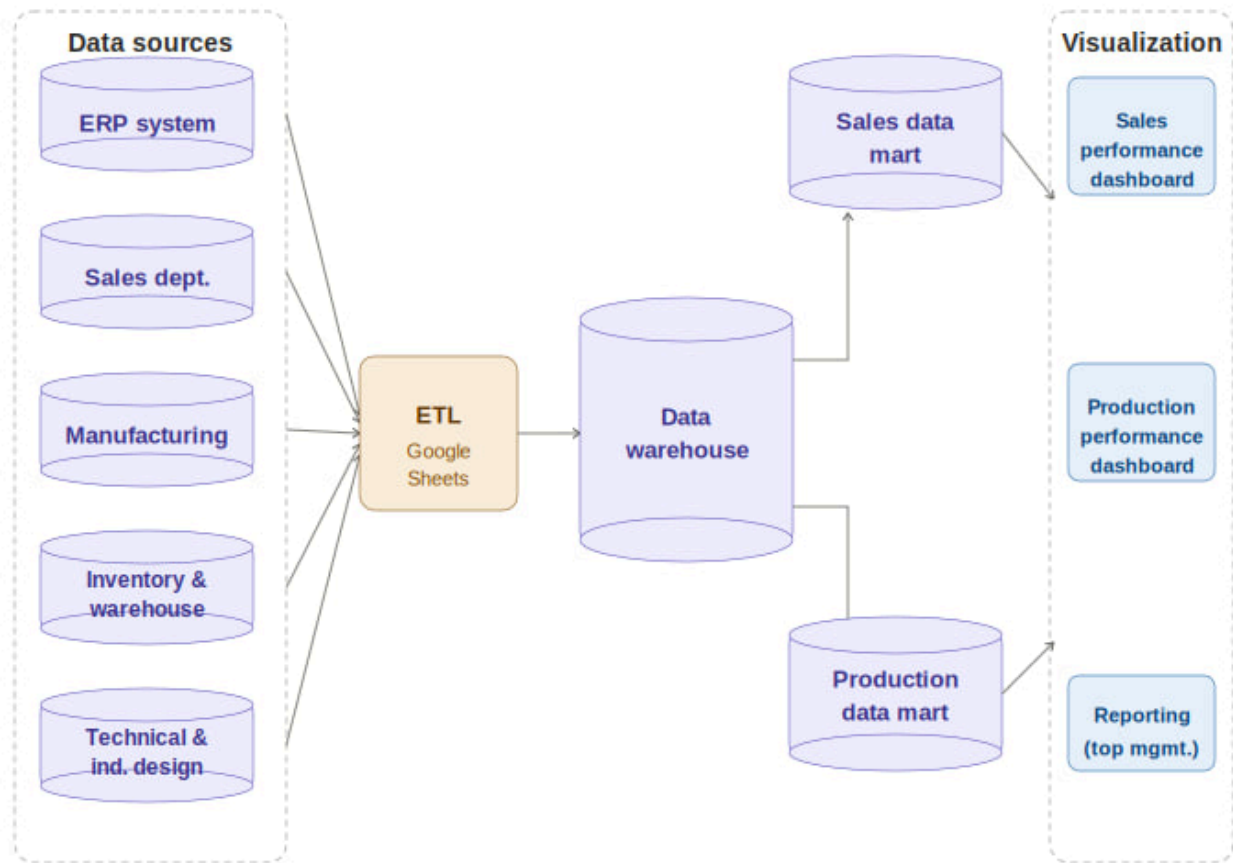


Figure 6.0 Data Architecture

6.1 Data Flow Summary

1. Raw operational data is captured by Manufacturing, Sales, Inventory, and Technical departments and fed into the ERP system.
2. Reports from ERP (production reports, sales reports) are exported and uploaded to Google Sheets.
3. The Data Analyst performs ETL: removes redundant data (cleansing), restructures tables, converts data types (e.g., string dates to date format).
4. The cleaned data warehouse in Google Sheets is connected to Google Data Studio via the native connector.
5. Google Data Studio reads the data and renders two interactive dashboards, accessible to Top Management via a web browser.

6. Management interacts with dashboards using filters (date range, product category, machine, customer ID) for self-service analysis.

7.0 Multidimensional Model

A galaxy schema is designed to support the OLAP (Online Analytical Processing) requirements of the BI dashboard. Two fact tables are defined, one for sales and one for production and each surrounded by dimension tables as shown in Figure 7.0.

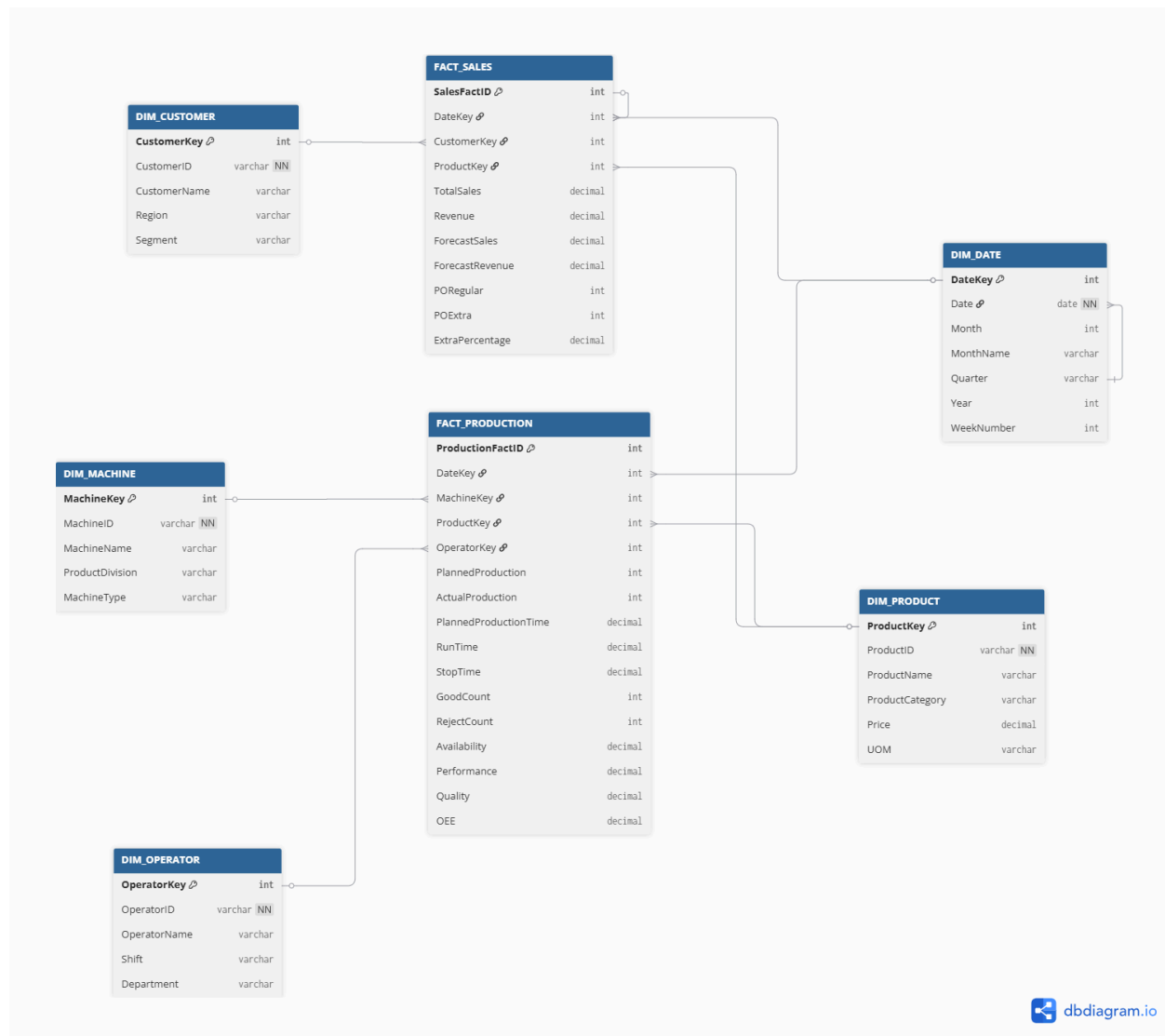


Figure 7.0 Galaxy Schema

8.0 Conclusion

This report analyzed the research paper on BI dashboard development at PT. MPM and addressed four key tutorial tasks. The main problem was the absence of a performance monitoring system, which impaired management decision-making. The solution involved developing two interactive BI dashboards (Sales and Production) using Google Data Studio under the Waterfall methodology, achieving an 82% user satisfaction score.

The ERD captures six key entities (Customer, Product, SalesTransaction, Machine, Operator, ProductionRecord) with clearly defined relationships. The Use Case Diagram defines six actor roles and twelve use cases governing system interactions. The system architecture follows a three-tier BI model: Data Sources → ETL (Google Sheets) → Presentation (Google Data Studio). Finally, the Star Schema defines two fact tables (FACT_SALES and FACT_PRODUCTION) supported by four dimension tables and a shared DIM_DATE conformed dimension, enabling multidimensional OLAP analysis across both operational domains.